Code No: 135AF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, July/August - 2023
DESIGN AND ANALYSIS OF ALGORITHMS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper July/Aug-2023

PART – A

(25 Marks)

1.a) Define θ-notation and Omega notation. [2]
b) How do you measure the algorithm running time? [3]
c) What is a set? List the operations that can be performed on it. [2]
d) State and explain graph coloring problem. [3]
e) Define greedy method. [2]
f) What are the drawbacks of greedy method? [3]
g) State the principle of optimality. [2]
h) Write about reliability designing. [3]
i) Define NP-Complete. [2]
j) Why the search path in a state-space tree of a branch and bound algorithm is terminated? [3]

PART – B

(50 Marks)

2.a) What is an algorithm? Explain its characteristics.
b) Derive the time complexity of quick sort in an average case. [5+5]

OR

3.a) Discuss about time complexity in detail.
b) Write the algorithm for merge sort with an example. [3+7]

4.a) Write an algorithm to determine the Hamiltonian cycle in a given graph using backtracking.
b) Discuss about AND/OR graph problem with an example. [5+5]

OR

5.a) Give the solution to the 8-queens problem using backtracking.
b) Write and explain the algorithm of Bi connected components with an example. [5+5]

6. Write and explain prim's algorithm for generating minimum cost Spanning tree and list some applications of it. [10]

OR

7.a) Explain the greedy technique for solving the Job Sequencing problem.
b) What are the applications of greedy method? Discuss. [5+5]

8.a) Explain the time and space complexity of Dynamic Programming traveling sales person algorithm.

b) Write and explain the algorithm of matrix chain multiplication. [5+5]

OR

9.a) Explain the all pairs shortest path problem with an example.

b) Write an algorithm of OBST. [5+5]

10.a) Show that the Hamiltonian path problem is NP-Complete.

b) Is the travelling salesman problem is NP-hard or NP-Complete? Justify your answer. [5+5]

OR

11.a) Write an algorithm of FIFO Branch and Bound.

b) Discuss in detail about Cook's theorem. [5+5]

---ooOoo---

used paper July/Aug-2023